TrafficEye: An Integrated Crowdsourced and AI-Assisted System for Traffic Violation Reporting and Enforcement

Meenakshi Garg[1], Shivam Chauhan[2], Mohit Jain[3], Aditya Jaiswar[4]

Vivekanand Education Society's Institute of Technology, Mumbai-400074, India

[1]meenakshi.garg@ves.ac.in

[2]2025.shivam.chauhan@ves.ac.in

[3]2025.mohit.jain@ves.ac.in

[4]2025.aditya.jaiswar@ves.ac.in

Abstract

Traffic violations often occur beyond the view of fixed cameras and routine police patrols, making citizen reports a useful additional source of evidence. The difficulty lies in turning an uploaded photograph or video into a report that an officer can review quickly. We developed TrafficEye as a prototype mobile-to-officer workflow for this purpose. The application captures or imports media, records GPS coordinates or available EXIF data, and stores the submission in a managed cloud backend. A pretrained Vision-Language Model (VLM) then proposes vehicle details, a violation category, license-plate text, and a severity label. These fields are advisory: the reviewing officer examines the original evidence and may confirm, correct, or reject the report. The contribution of this work is the integration of these functions rather than a new detection algorithm. Quantitative testing is still pending, so we do not claim improvements in recognition accuracy or review time. The planned evaluation will measure violation classification, vehicle-detection recall, plate-reading accuracy, and severity agreement under both clear and difficult capture conditions.

Keywords: *Traffic Violation Reporting · Citizen Crowdsourcing · Vision-Language Model (VLM) · Multimodal Visual Analysis · Automatic Number Plate Recognition · GPS and EXIF Geolocation · Officer Verification*

1. Introduction

Fixed cameras and police patrols cannot observe every road at all times. Citizen reporting can add evidence from places and periods that formal surveillance misses [1-4]. Previous systems show that road users can document hazardous locations, traffic offences, and other safety concerns with mobile phones [2-5]. In India, Delhi Traffic Prahari already accepts photographs and videos for police verification [6]. Collecting the media, however, is only the first step. A reviewer must still identify the vehicle, interpret the event, read the registration plate, and decide whether the case warrants attention.

Automatic number-plate recognition and helmet-detection systems can automate parts of this work, but they are usually developed for a narrow task and depend on task-specific data or models [7-10]. TrafficEye takes a systems approach. It combines citizen submission with a pretrained VLM that proposes structured fields for the officer reviewing the case. The project does not develop or train a new vision model. Its contribution lies in the reporting workflow and in the way the mobile client, cloud services,

visual inference, and officer interface operate together. Accuracy and operational value must therefore be established by testing the complete application.

2. Related Work

Research on crowdsourced traffic enforcement raises a practical question: how should authorities separate usable evidence from irrelevant or poor-quality submissions? RoadCop uses a bi-modular process that first filters citizen videos for spam and then evaluates the remaining material for enforcement [1]. This separation is relevant to TrafficEye, although TrafficEye covers an additional stage. A submission that passes basic screening may still require plate transcription and violation classification. Automated suggestions therefore support the officer's review instead of replacing it.

CrowdOut supports real-time offence reporting and map-based visualization [2]. Mobile Roadwatch also links smartphone video with time and location tags and allows users to select events for police reporting [12]. Both systems demonstrate the value of attaching spatial context to citizen media. Their published descriptions, however, do not establish that they perform number-plate recognition or assign severity. TrafficEye treats these functions separately: location metadata records where the event was captured, while visual inference proposes what the evidence may show.

Other studies use citizen contributions to identify broader road-safety problems. Olma et al. analyse road users' reports of danger spots, while Dunlop et al. examine smartphone reports used to locate dangerous road sections and provide warnings [3, 4]. These applications combine observations about places. An individual violation report has a narrower evidentiary purpose and must preserve the relationship between the media, location, vehicle information, and review outcome. TrafficEye is organized around that single case record.

Indian traffic-police applications provide a relevant operational model. Traffic Prahari asks users to submit media together with the date, time, location, registration number, and violation type before police verification [6]. The Mumbai Traffic Police application also accepts citizen reports [13]. These examples show that citizen-to-authority reporting is practical. The cited public descriptions do not document the full extent of their visual analysis or backend behaviour, so this paper does not infer unstated features. TrafficEye retains the police-review step and uses model-produced fields to reduce the information that must be entered during initial triage.

Task-specific computer vision offers greater control when a system has a clearly defined target. Srivastava et al. combine YOLOv5 with EasyOCR for Indian number plates [7], whereas Deshpande et al. use a two-stage method for helmet-violation detection and vehicle identification [8]. Their work provides useful evaluation references for the plate and violation tasks considered separately. It also illustrates the engineering effort required to maintain several specialized pipelines. TrafficEye instead calls a hosted multimodal model for multiple fields. This simplifies application development, but it gives the project less control over each detection stage and still requires application-specific validation.

Reviews of automatic number-plate recognition describe plate reading as a sequence of detection, recognition, OCR, and image-processing decisions [9, 10]. Performance varies with lighting, camera angle, weather, plate condition, and the dataset used for testing. A plausible string is therefore not sufficient evidence that TrafficEye reads plates reliably. The proposed evaluation will report exact plate matches and character-level accuracy, including difficult captures. Results from dedicated ANPR systems

provide methodological context, but their accuracy figures cannot be transferred directly to this application.

Vision-language models offer a reusable option when one application needs several related interpretations from the same image. Zhang et al. survey models that connect visual and language representations [15], and Caffagni et al. review multimodal capabilities, datasets, and benchmarks [16]. TrafficEye uses an image-understanding API from this model class [14]. These studies support investigating a pretrained multimodal component, but they do not establish its accuracy on TrafficEye's evidence. The application must test whether one fixed prompt can produce dependable vehicle descriptions, violation labels, plate text, and severity suggestions.

Technical accuracy is only one condition for adoption. Khojastehpour et al. report that privacy, perceived fairness of penalties, and trust in police performance influence acceptance of crowdsourced enforcement [5]. Dasgupta et al. also stress human involvement and accountability in AI-supported law enforcement [11]. TrafficEye therefore leaves the enforcement decision with the officer and identifies bystander anonymization as necessary for future work. A severity label may help order the review queue, but the original evidence remains the basis of the decision.

Prior work covers the main parts of the problem, usually for different purposes: collecting citizen evidence, mapping locations, recognizing specific visual features, or supporting accountable review [1-5, 7-9, 12]. TrafficEye brings these functions into one submission record. GPS and EXIF data provide location context; cloud storage and a relational database keep the evidence and case data together; the VLM supplies candidate fields; and the officer records the final decision. The value of this integration has not yet been measured against a manual reporting workflow.

3. Problem Statement

TrafficEye addresses five related operational problems:

- **Limited enforcement coverage:** fixed cameras and patrols cannot observe every road continuously.
- **Inconsistent citizen evidence:** media varies in clarity, lighting, viewpoint, and completeness.
- **Time-consuming manual triage:** officers must identify vehicles, interpret violations, and transcribe plates.
- **Uncertain location metadata:** GPS and EXIF may be missing or altered and cannot serve as proof alone.
- **Accountable automation:** AI may assist review, but final decisions must remain with authorized officers.

The system must therefore connect evidence capture, location context, structured candidate fields, and officer verification without treating model output as a final decision.

4. TrafficEye System and Methodology

TrafficEye uses a short Capture, Verify, Submit flow. A citizen takes a photograph or video, or selects existing media, then checks the detected location and the fields prepared by the application before submitting the report. This follows the smartphone participation model described in earlier road-safety systems [3, 4]. The backend analyses the evidence and adds candidate fields before the case appears in the officer queue. During review, the officer sees the original media, corrects or confirms the proposed information, and accepts or rejects the case. The citizen then receives the updated status. This division keeps the model in an assistive role and preserves officer responsibility [11].

The prototype has not yet undergone quantitative evaluation. The planned study will use approximately 50 to 100 traffic images and short video clips covering supported offences, including no-helmet, red-light,

and wrong-parking violations. The sample will contain clear captures as well as low-light scenes, oblique views, and partly hidden plates. It will be used only for evaluation; the project will not train the model on these items. General multimodal benchmarks cannot show how the complete reporting workflow performs, so the measurements must be collected within TrafficEye [15, 16].

Each item will be labelled independently by the three project members for vehicle presence, violation type, correct plate text, and severity. Disagreements will be discussed until a reference label is agreed. TrafficEye will then process the same items with an unchanged model and prompt configuration. The analysis will report violation-classification accuracy, vehicle-detection recall, exact plate matches, character-level OCR accuracy, and agreement between the proposed and reference severity labels. Because the application uses a hosted pretrained model, the protocol contains no training or fine-tuning stage.

5. System Architecture

The system connects the citizen application and the Traffic Officer Command Center through a managed cloud backend. As shown in Figure 1, a submission contains the captured media and available location metadata. The backend stores that material, records the report in a relational database, and synchronizes status changes with the officer interface. A separate API request sends the evidence to the VLM, which returns proposed vehicle details, violation labels, plate text, and severity [14-16]. These outputs are stored with the report but remain subject to officer review.

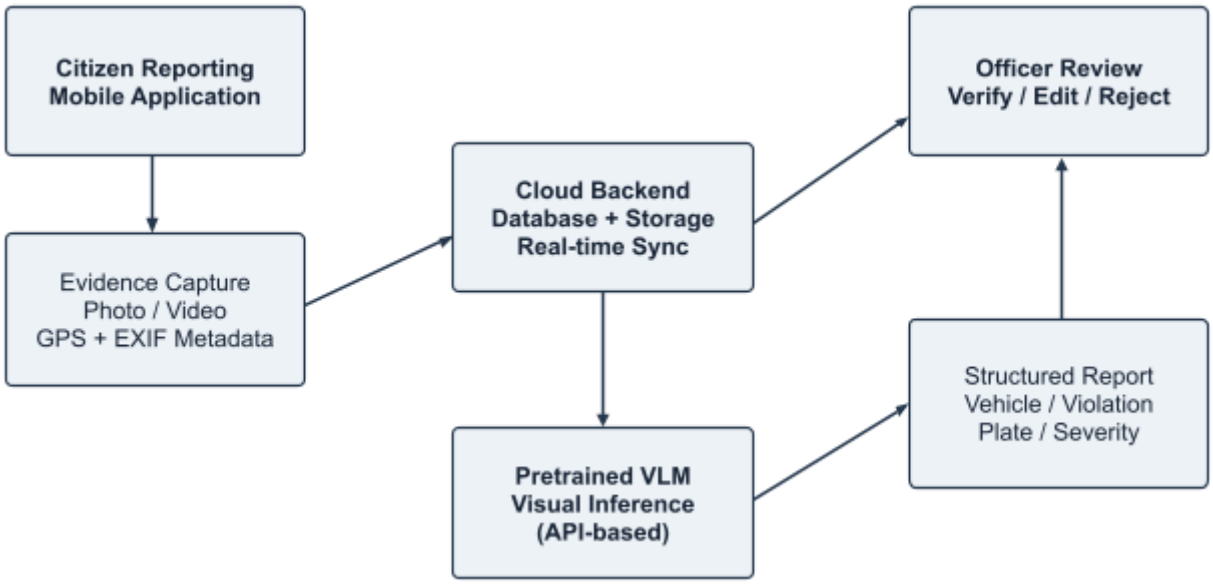


Fig. 1. TrafficEye system architecture: citizen capture, cloud backend, VLM inference, and officer review.

After upload, the backend keeps the original evidence and metadata together while the inference service produces structured candidate fields. The officer queue presents both sources so that reviewers can compare the model output with the submitted media. Figure 1 separates the storage path from the analysis path and shows where they meet in the structured report. Severity may influence queue order, but it does not determine the outcome. Confirmations, corrections, and rejections are written back to the shared report and shown to the citizen.

6. Implementation

We built the mobile client from a shared Android and iOS codebase. Its managed development toolkit provides access to the camera, device location, and local storage. For a photograph taken inside the application, the client records the device's current latitude and longitude. For imported media, it reads available EXIF coordinates, capture time, and device fields, then compares the stored coordinates with the user's current position. This comparison is a plausibility check rather than proof of origin because EXIF fields may be edited, removed, or absent. The interface follows the project's navy-and-amber visual theme.

The backend stores submitted media in object storage and keeps report fields in a managed relational database. Real-time subscriptions notify the officer interface when a new report or status change is available, avoiding a manual refresh. Each database record links the evidence and location data to the candidate analysis fields and review status, and officers can query the queue by status, location, or severity. The design anticipates concurrent submissions, but formal load tests have not established its throughput or scaling behaviour. Upload and remote inference also depend on network access, so an interrupted connection can delay or prevent submission.

Before sending evidence to the model, the processing workflow compresses the media and converts it to a supported format. The hosted VLM receives the image or selected video frames with a task-specific prompt that requests structured fields. The application parses the response into a vehicle description, possible violation, plate text, and one of four severity levels: Low, Medium, High, or Critical [14-16]. If the first response does not provide usable plate text, the application can make a follow-up inference request. Figure 2 summarizes the sequence from preprocessing to officer review.

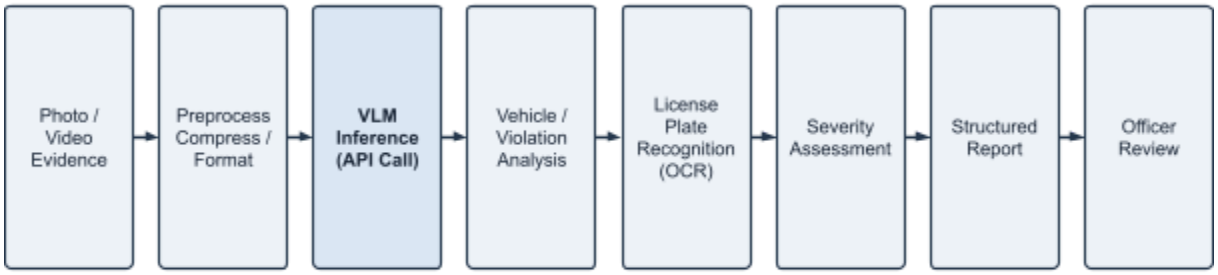


Fig. 2. Conceptual VLM inference pipeline used by TrafficEye (inference only; no training/fine-tuning).

TrafficEye uses the pretrained model only for inference. It does not modify the model, build a training dataset, or divide project data into training, validation, and test sets. This reduces the engineering required for model development, although it also gives the team less control than a dedicated detector would provide [7, 8]. Once the response has been parsed, the officer can inspect the original media, amend the candidate fields, and record a decision. The application was designed to reduce repetitive data entry during triage, but improvements in review time and reporting completion have not yet been measured.

7. Prototype Results and Limitations

At this stage, the main result is a working prototype. It captures or imports citizen evidence, attaches GPS or EXIF context, synchronizes the report through the cloud backend, and presents model-generated fields to the reviewing officer. Figures 3 and 4 show the pending queue and the review screen. These interfaces show the implemented workflow, but they do not demonstrate detection accuracy or operational improvement. Application-level testing is still pending; consequently, the paper does not report classification accuracy, vehicle-detection recall, plate-recognition accuracy, or severity agreement. Published multimodal benchmarks cannot substitute for measurements on the project's traffic cases [15, 16]. The same limitation applies to claims about faster review or better user experience, neither of which has been measured. Potential failure cases include low light, heavy rain, obstructed plates, and damaged plates [10, 15, 16]. Editable location metadata, network dependence, and the lack of automatic bystander anonymization create further limits. These issues justify officer verification and must be included in later evaluation of privacy, fairness, and trust [5, 11].

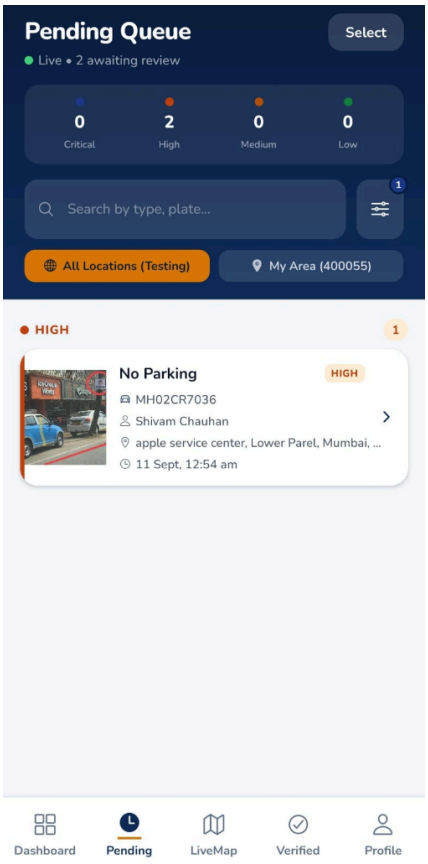


Fig. 3. Officer pending-review queue with severity-based triage, real-time sync, and location filtering.

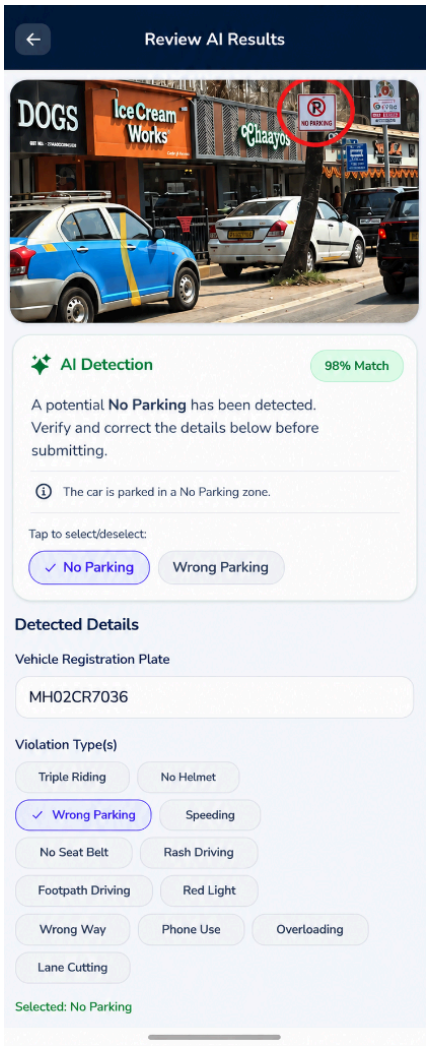


Fig. 4. AI-assisted review: 98% confidence VLM detection of a No Parking violation with extracted plate and editable fields.

8. Conclusion and Future Work

TrafficEye combines mobile crowdsourcing, location metadata, cloud services, and pretrained visual inference within a citizen-to-officer reporting workflow. Its main contribution is the integration of these components with human verification rather than a new detection algorithm. The prototype demonstrates a practical reporting and review process, although its accuracy and operational value still require formal evaluation. The immediate priority is to conduct the planned domain-specific study and examine plate recognition across varied lighting, viewpoints, and levels of occlusion.

Further development should address the limits observed in the current design. Officer queues could be filtered by precinct or distance, and a city heatmap could support patrol planning. Bystanders and uninvolved vehicles should be anonymized before evidence is displayed outside the immediate review context. The backend requires load testing, and location checking should combine EXIF data with independent signals before any confidence score is treated as evidentiary. Later versions may connect verified cases to authorized registration lookup, digital challan, and vehicle-owner notification services through an approved RTO or DMV interface. None of these features has yet been implemented or evaluated.

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Disclosure of Interests

The authors declare no competing interests related to this work.

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